

Payment Ecosystem IMTO-CRH

Base URL: <https://mule.ecobank.com/pecrh>

Encryption/Decryption Flow

- i. AES key is generated and used to encrypt payload using the AES/GCM/NoPadding Cipher
- ii. The Generated Key is then Encrypted using the other server’s RSA Public Key
- iii. The encrypted key is passed in the x-secret-key header
- iv. The server reads the header value and decrypts using its private key
- v. The plain key is used to decrypt the payload and payload is processed
- vi. Steps 1-3 are repeated for the response back to the client

Authentication

Service Name	Authenticate		
Method	POST		
URL	/authenticate		
Field	Type	M/O	Description
username	ALPHANUMERIC	M	Username Provided to Client
password	ALPHANUMERIC	M	Password Provided to Client
Sample Request & Response JSON			
Request	<pre>{ "username": "isw", "password": "XY7kmzoNzl100" }</pre>		
Response	<pre>{ "token": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJJU1ciLCJleHAiOiJlE2MTQ5MTk1MTgsImhhdCI6MTYxNDkwMTUxOH0.xJz0rhZyxP6rVA3DtafadnCbtyvYfs2TobjwCX__InFILOq-3hULR3TAxDM9dgMLNh2_LfhzKjoe7rI1qfgIVg" }</pre>		

Mandatory Headers

Header	Header Description
Authentication	Authentication Token in the format “Bearer {{token}}”
x-secret-key	RSA Encrypted AES Key Value

Fetch Institution List

Service Name	Fetch Institution List		
Method	POST		
URL	/imto/fetchinstitutions		
Field	Type	M/O	Description
country	ALPHANUMERIC	M	2 Digit Country Code e.g., “NG”
Sample Request & Response JSON			
Request	<pre>{ "country": "NG" }</pre>		
Response	<pre>{ "responseCode": "000", "responseMessage": "TRANSACTION PROCESSED SUCCESSFULLY", "institutionListList": [{ "institutionId": "3LINECML-ACC", "institutionName": "3LINE CARD MANAGEMENT LIMITED", "institutionType": "BANK", "countryCode": "ENG" }, { "institutionId": "ABMFB-ACC", "institutionName": "AB MICROFINANCE BANK", "institutionType": "BANK", "countryCode": "ENG" }] }</pre>		

Account Enquiry (Ecobank, Other Banks)

Service Name	Account Enquiry		
Method	POST		
URL	/imto/accountenquiry		
Field	Type	M/O	Description
accountNo	ALPHANUMERIC	M	Account Number to Validate
destinationEntityCode	ALPHANUMERIC	M	Destination Entity Code ("ECOBANK"/ Institution ID from Fetch Institution List)
deliveryMethod	ALPHANUMERIC	M	Delivery Method (ACCOUNT/CASH/WALLET/INTER BANK)
Sample Request & Response JSON			
Request	<pre>{ "accountNo": "0023123747", "destinationEntityCode": "ECOBANK", "deliveryMethod": "ACCOUNT", "destinationCountry": "NG" }</pre>		
Response	<pre>{ "responseCode": "000", "responseMessage": "Success", "accountNo": "0023123747", "accountName": "KASUMU GAFFAR OLANREWAJU", "ccy": "NGN", "branchCode": "C35", "accountType": "U", "accountClass": "GHCATF", "accountStatus": "ACTIVE", "firstName": "Gaffar", "lastName": "Kasumu", "gender": "M", "idType": "INTL_PASSPORT", "idNumber": "A234578689", "address": "1, Ahmadu Bello", "nationality": "Nigeria", "email": "GAFFYLISE@gmail.com", }</pre>		

	"phoneNumber":"2347062725721"
	}

To Account Transfer

Service Name	Account to Account Post Transfer		
Method	POST		
URL	/imto/account/process		
Field	Type	M/O	Description
amount	NUMERIC	M	Settlement Amount
receiverAmount	Numeric	M	Receive Amount
receiverCcy	ALPHANUMERIC	M	Receive Currency
destinationCountry	ALPHANUMERIC	M	2 Digit Destination Country Code e.g., “NG”
narration	ALPHANUMERIC	M	Narration
purpose	ALPHANUMERIC	M	Purpose
receiverFirstName	ALPHANUMERIC	M	Receiver First Name
receiverLastName	ALPHANUMERIC	M	Receiver Last Name
receiverNationality	ALPHANUMERIC	O	Receiver Nationality
senderFirstName	ALPHANUMERIC	M	Sender First Name
senderLastName	ALPHANUMERIC	M	Sender Last Name
senderNationality	ALPHANUMERIC	M	Sender Nationality
sourceCountry	ALPHANUMERIC	M	2 Digit Source Country Code e.g., “NG”
transactionReference	ALPHANUMERIC	M	Unique Transaction Reference
senderResidentialAddress	ALPHANUMERIC	M	Sender Residential Address
receiverResidentialAddress	ALPHANUMERIC	O	Receiver Residential Address
exchangeRate	NUMERIC	M	Exchange Rate
destinationAccountNumber	NUMERIC	M	Destination Account Number
destinationBank	ALPHANUMERIC	M	Destination Bank, “ECOBANK” in this case
List<UDFData>	OBJECT	O	User Defined Data Type for Extra Customizations

udfName	ALPHANUMERIC	O	
udfValue	ALPHANUMERIC	O	
udfDataType	ALPHANUMERIC	O	
udfCode	ALPHANUMERIC	O	
Sample Request & Response JSON			
Request	<pre>{ "amount": "20", "receiverCcy": "USD", "receiverAmount": "20", "destinationCountry": "NG", "narration": "Test Transaction", "purpose": "Travel", "receiverFirstName": "Gaffar", "receiverLastName": "Kasumu", "receiverNationality": "Nigeria", "senderFirstName": "James", "senderLastName": "Oladimeji", "senderNationality": "France", "sourceCountry": "NG", "transactionReference": "ISWXX12371XXX", "senderResidentialAddress": "4, Iganmu Road", "receiverResidentialAddress": "8, Accra Level", "exchangeRate": "1", "destinationAccountNumber": "0010045373", "destinationBank": "ECOBANK" }</pre>		
Response	<pre>{ "responseCode": "001", "responseMessage": "PENDING", "externalReference": "ISWXX12367XXX", "cbaReference": "ISWXX12367XXX" }</pre>		

To Cash Transfer

Service Name	To Cash Post Transfer		
Method	POST		
URL	/imto/cash/process		
Field	Type	M/O	Description
amount	NUMERIC	M	Settlement Amount

beneficiaryAmount	Numeric	M	Beneficiary Receive Amount
destinationCountry	ALPHANUMERIC	M	2 Digit Destination Country Code e.g., “NG”
narration	ALPHANUMERIC	M	Narration
purpose	ALPHANUMERIC	M	Purpose
receiverFirstName	ALPHANUMERIC	M	Receiver First Name
receiverLastName	ALPHANUMERIC	M	Receiver Last Name
receiverNationality	ALPHANUMERIC	M	Receiver Nationality
senderFirstName	ALPHANUMERIC	M	Sender First Name
senderLastName	ALPHANUMERIC	M	Sender Last Name
senderNationality	ALPHANUMERIC	M	Sender Nationality
sourceCountry	ALPHANUMERIC	M	2 Digit Source Country Code e.g., “NG”
receiverIdType	ALPHANUMERIC	M	Receiver ID Type
receiverIdNumber	ALPHANUMERIC	M	Receiver ID Number
receiverPhoneNumber	ALPHANUMERIC	O	Receiver Phone Number
senderIdType	ALPHANUMERIC	M	Sender ID Type
senderDOB	ALPHANUMERIC	M	Sender Date of Birth e.g., “1994-10-29”
receiverDOB	ALPHANUMERIC	M	Receiver Date of Birth e.g., “1994-10-29”
senderIdNumber	ALPHANUMERIC	M	Sender ID Number
senderIdExpiryDate	ALPHANUMERIC	M	Sender ID Expiry Date
senderIdIssueDate	ALPHANUMERIC	M	Sender ID Card Issue Date
senderPhoneNumber	ALPHANUMERIC	O	Sender Phone Number
receiverCcy	ALPHANUMERIC	M	Receiver Currency
transactionReference	ALPHANUMERIC	M	Unique Transaction Reference
senderResidentialAddress	ALPHANUMERIC	M	Sender Residential Address
receiverResidentialAddress	ALPHANUMERIC	M	Receiver Residential Address
exchangeRate	NUMERIC	M	Exchange Rate
destinationAccountNumber	NUMERIC	M	Destination Account Number
destinationBank	ALPHANUMERIC	M	Destination Bank, “ECOBANK” in this case
testAnswer	ALPHANUMERIC	M	Test Question to Pick Up Cash
testQuestion	ALPHANUMERIC	M	Test Answer to Pick Up Cash
List<UDFData>	OBJECT	O	

			User Defined Data Type for Extra Customizations
udfName	ALPHANUMERIC	O	
udfValue	ALPHANUMERIC	O	
udfDataType	ALPHANUMERIC	O	
udfCode	ALPHANUMERIC	O	
Sample Request & Response JSON			
Request	<pre>{ "amount": "20", "amountType": "SENDAMT", "destinationCountry": "NG", "narration": "Test Transaction", "purpose": "Travel", "receiverFirstName": "Gaffar", "receiverLastName": "Kasumu", "receiverNationality": "Nigeria", "senderFirstName": "James", "senderLastName": "Oladimeji", "senderNationality": "France", "sourceCountry": "NG", "receiverIdType": "INTL_PASSPORT", "receiverIdNumber": "124354627", "receiverPhoneNumber": "2347062725721", "senderIdType": "INTL_PASSPORT", "senderDOB": "1994-10-29", "receiverDOB": "1994-10-29", "senderIdNumber": "09838773822", "senderIdExpiryDate": "2022-10-29", "senderIdIssueDate": "2019-10-29", "transactionReference": "ISWXX0001XXCH", "exchangeRate": "1", "receiverCcy": "USD", "senderResidentialAddress": "4, Iganmu Road", "receiverResidentialAddress": "8, Accra Level", "testAnswer": "What is the name of your pet ?", "testQuestion": "Jerry" }</pre>		
Response	<pre>{ "responseCode": "001", "responseMessage": "PENDING", }</pre>		

	<pre>"externalReference": "ISWXX0001XXCH", "cbaReference": "ISWXX0001XXCH" }</pre>
--	--

To Account (Interbank)

Service Name	Account to Account (Interbank) Transfer		
Method	POST		
URL	/imto/interbank/process		
Field	Type	M/O	Description
destinationAccountNumber	NUMERIC	M	Destination Account Number
destinationBank	NUMERIC	M	Destination Bank Code (From Fetch Institution List)
amount	NUMERIC	M	Settlement Amount
receiverAmount	NUMERIC	M	Receive Amount
receiverCcy	ALPHANUMERIC	M	Receive Currency
destinationCountry	ALPHANUMERIC	M	2 Digit Destination Country Code e.g., "NG"
narration	ALPHANUMERIC	M	Narration
purpose	ALPHANUMERIC	M	Purpose
receiverFirstName	ALPHANUMERIC	O	Receiver First Name
receiverLastName	ALPHANUMERIC	O	Receiver Last Name
receiverNationality	ALPHANUMERIC	O	Receiver Nationality
senderFirstName	ALPHANUMERIC	M	Sender First Name
senderLastName	ALPHANUMERIC	M	Sender Last Name
senderNationality	ALPHANUMERIC	M	Sender Nationality
sourceCountry	ALPHANUMERIC	M	2 Digit Source Country Code e.g., "NG"
receiverIdType	ALPHANUMERIC	O	Receiver ID Type
receiverIdNumber	ALPHANUMERIC	O	Receiver ID Number
receiverPhoneNumber	ALPHANUMERIC	O	Receiver Phone Number
senderIdType	ALPHANUMERIC	M	Sender ID Type
senderDOB	ALPHANUMERIC	M	Sender Date of Birth e.g., "1994-10-29"
receiverDOB	ALPHANUMERIC	O	Receiver Date of Birth e.g., "1994-10-29"
senderIdNumber	ALPHANUMERIC	M	Sender ID Number
senderIdExpiryDate	ALPHANUMERIC	M	Sender ID Expiry Date

senderIdIssueDate	ALPHANUMERIC	M	Sender ID Card Issue Date
senderPhoneNumber	ALPHANUMERIC	O	Sender Phone Number
transactionReference	ALPHANUMERIC	M	Unique Transaction Reference
senderResidentialAddress	ALPHANUMERIC	M	Sender Residential Address
receiverResidentialAddress	ALPHANUMERIC	O	Receiver Residential Address
exchangeRate	NUMERIC	M	Exchange Rate
List<UDFData>	OBJECT	O	User Defined Data Type for Extra Customizations
udfName	ALPHANUMERIC	O	
udfValue	ALPHANUMERIC	O	
udfDataType	ALPHANUMERIC	O	
udfCode	ALPHANUMERIC	O	

Sample Request & Response JSON

Request	<pre>{ "destinationAccountNumber":"0035164343", "destinationBank":"300323", "amount":"20", "receiverAmount":"100", "destinationCountry":"GH", "narration":"Test Transaction", "purpose":"Travel", "receiverFirstName":"Gaffar", "receiverLastName":"Kasumu", "receiverNationality":"Nigeria", "senderFirstName":"James", "senderLastName":"Oladimeji", "senderNationality":"France", "sourceCountry":"NG", "receiverIdType":"INTL_PASSPORT", "receiverIdNumber":"124354627", "receiverPhoneNumber":"2347062725721", "senderIdType":"INTL_PASSPORT", "senderDOB":"1994-10-29", "receiverDOB":"1994-10-29", "senderIdNumber":"09838773822", "senderIdExpiryDate":"2022-10-29", "senderIdIssueDate":"2019-10-29", "transactionReference":"ISWXX0001XXIB",</pre>
---------	--

	<pre>"exchangeRate":"5.00", "receiverCcy":"GHS", "senderResidentialAddress":"4, Iganmu Road", "receiverResidentialAddress":"8, Accra Level " }</pre>
Response	<pre>{ "responseCode": "001", "responseMessage": "PENDING", "externalReference": " ISWXX0001XXIB", "cbaReference": " ISWXX0001XXIB" }</pre>

TO WALLET

Service Name	Account to Wallet Transfer		
Method	POST		
URL	/imto/wallet/process		
Field	Type	M/O	Description
destinationAccountNumber	NUMERIC	M	Destination Account Number
destinationBank	NUMERIC	M	Destination Bank Code (From Fetch Institution List) e.g., "VODAFONEGH"
amount	NUMERIC	M	Settlement Amount
receiverAmount	Numeric	M	Receive Amount
destinationCountry	ALPHANUMERIC	M	2 Digit Destination Country Code e.g., "NG"
narration	ALPHANUMERIC	M	Narration
purpose	ALPHANUMERIC	M	Purpose
receiverFirstName	ALPHANUMERIC	M	Receiver First Name
receiverLastName	ALPHANUMERIC	M	Receiver Last Name
receiverNationality	ALPHANUMERIC	O	Receiver Nationality
senderFirstName	ALPHANUMERIC	M	Sender First Name
senderLastName	ALPHANUMERIC	M	Sender Last Name
senderNationality	ALPHANUMERIC	M	Sender Nationality
sourceCountry	ALPHANUMERIC	M	2 Digit Source Country Code e.g., "NG"
receiverIdType	ALPHANUMERIC	O	Receiver ID Type
receiverIdNumber	ALPHANUMERIC	O	Receiver ID Number

receiverPhoneNumber	ALPHANUMERIC	O	Receiver Phone Number
senderIdType	ALPHANUMERIC	M	Sender ID Type
senderDOB	ALPHANUMERIC	M	Sender Date of Birth e.g., "1994-10-29"
receiverDOB	ALPHANUMERIC	O	Receiver Date of Birth e.g., "1994-10-29"
senderIdNumber	ALPHANUMERIC	M	Sender ID Number
senderIdExpiryDate	ALPHANUMERIC	M	Sender ID Expiry Date
senderIdIssueDate	ALPHANUMERIC	M	Sender ID Card Issue Date
senderPhoneNumber	ALPHANUMERIC	O	Sender Phone Number
receiverCcy	ALPHANUMERIC	M	Receiver Currency
transactionReference	ALPHANUMERIC	M	Unique Transaction Reference
senderResidentialAddress	ALPHANUMERIC	M	Sender Residential Address
receiverResidentialAddress	ALPHANUMERIC	O	Receiver Residential Address
exchangeRate	NUMERIC	M	Exchange Rate
List<UDFData>	OBJECT	O	User Defined Data Type for Extra Customizations
udfName	ALPHANUMERIC	O	
udfValue	ALPHANUMERIC	O	
udfDataType	ALPHANUMERIC	O	
Sample Request & Response JSON			
Request	<pre>{ "destinationAccountNumber":"0035164343", "destinationBank":"VODAFONEGH", "amount":"20", "receiverAmount":"100", "destinationCountry":"GH", "narration":"Test Transaction", "purpose":"Travel", "receiverFirstName":"Gaffar", "receiverLastName":"Kasumu", "receiverNationality":"Nigeria", "senderFirstName":"James", "senderLastName":"Oladimeji", "senderNationality":"France", "sourceCountry":"ES", "receiverIdType":"INTL_PASSPORT", }</pre>		

	<pre>"receiverIdNumber":"124354627", "receiverPhoneNumber":"2347062725721", "senderIdType":"INTL_PASSPORT", "senderDOB":"1994-10-29", "receiverDOB":"1994-10-29", "senderIdNumber":"09838773822", "senderIdExpiryDate":"2022-10-29", "senderIdIssueDate":"2019-10-29", "transactionReference":"ISWXX0001XXWT", "exchangeRate":"5.00", "receiverCcy":"GHS", "senderResidentialAddress":"4, Iganmu Road", "receiverResidentialAddress":"8, Accra Level " }</pre>
Response	<pre>{ "responseCode": "001", "responseMessage": "PENDING", "externalReference": " ISWXX0001XXWT", "cbaReference": " ISWXX0001XXWT" }</pre>

CALLBACK API (CLIENT SIDE)

Service Name	Client Call Back		
Method	POST		
URL	{to be provided by you...}		
Field	Type	M/O	Description
transactionReference	ALPHANUMERIC	M	Transaction Reference
responseCode	ALPHANUMERIC	M	Transaction Response Code
responseMessage	ALPHANUMERIC	M	Transaction Response Message
Sample Request & Response JSON			
Request	<pre>{ "transactionReference":"ISWXX0001XXWT", "responseCode":"000", "responseMessage":"Success" }</pre>		
Response	<pre>{ "responseCode":"000",</pre>		

	<pre>"responseMessage": "Success" }</pre>
--	---

Transaction Re-Query

Service Name	Transaction Enquiry End Point		
Method	POST		
URL	/imto/transactionenquiry		
Field	Type	M/O	Description
transactionReference	ALPHANUMERIC	M	Transaction Reference
Sample Request & Response JSON			
Request	<pre>{ "transactionReference": "ISWXX0001XXWT" }</pre>		
Response	<pre>{ "responseCode": "000", "responseMessage": "Success", "transactionReference": "ISWXX0001XXWT", "cbaReference": "1234567" }</pre>		

Response Code	Response Message
SUCCESS	000
ERROR	999
TRANSACTION_REF_EXISTS	E01
TRANSACTION_SAVE_FAILED	E02
TRANSACTION_STILL_PROCESSING	E00
TRANSACTION_PENDING_COMPLIANCE	E08
INVALID_RECEIVE_CURRENCY	E06
INVALID_BENEFICIARY_ACCOUNT	E04
MISSING_FIELD_IN_REQUEST	E05
TRANSACTION_REF_NOT_FOUND	E07